

Copula Entropy

Theory and Applications

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- Lecture 7: Generalizations
- **Lecture 8: Real applications and softwares**

Lecture 8: Real applications and softwares

1 Real applications

- Astronomy
- Physics
- Chemistry
- Geology
- Biology
- Medicine
- Social sciences
- Engineering

2 Softwares

- Official
- Third-party

¹Jian Ma. "Photometric Redshifts with Copula Entropy". In: *arXiv preprint arXiv:2310.16633* (2023).

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180	181	182	183	184	185	186	187	188	189	190	191	192	193	194	195	196	197	198	199	200	201	202	203	204	205	206	207	208	209	210	211	212	213	214	215	216	217	218	219	220	221	222	223	224	225	226	227	228	229	230	231	232	233	234	235	236	237	238	239	240	241	242	243	244	245	246	247	248	249	250	251	252	253	254	255	256	257	258	259	260	261	262	263	264	265	266	267	268	269	270	271	272	273	274	275	276	277	278	279	280	281	282	283	284	285	286	287	288	289	290	291	292	293	294	295	296	297	298	299	300	301	302	303	304	305	306	307	308	309	310	311	312	313	314	315	316	317	318	319	320	321	322	323	324	325	326	327	328	329	330	331	332	333	334	335	336	337	338	339	340	341	342	343	344	345	346	347	348	349	350	351	352	353	354	355	356	357	358	359	360	361	362	363	364	365	366	367	368	369	370	371	372	373	374	375	376	377	378	379	380	381	382	383	384	385	386	387	388	389	390	391	392	393	394	395	396	397	398	399	400	401	402	403	404	405	406	407	408	409	410	411	412	413	414	415	416	417	418	419	420	421	422	423	424	425	426	427	428	429	430	431	432	433	434	435	436	437	438	439	440	441	442	443	444	445	446	447	448	449	450	451	452	453	454	455	456	457	458	459	460	461	462	463	464	465	466	467	468	469	470	471	472	473	474	475	476	477	478	479	480	481	482	483	484	485	486	487	488	489	490	491	492	493	494	495	496	497	498	499	500	501	502	503	504	505	506	507	508	509	510	511	512	513	514	515	516	517	518	519	520	521	522	523	52
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热学

- 微热管设计和制造
 - 提出了一种基于 CE 和神经网络模型相结合的微热管设计和制造参数预测方法⁶
- 循环流化床锅炉工艺参数优化
 - 结合 CE、加权融合树模型和多目标树结构估计器的循环流化床锅炉工艺参数优化方法⁷

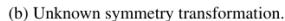
⁶ 李勇 et al. “基于一维卷积残差网络的微热管几何结构及制造工艺参数预测”. In: 中国科学: 技术科学 55.2 (2025), pp. 281–294, 张靖昊. “基于机器学习的微热管结构及工艺参数设计系统开发”. 硕士学位论文. 华南理工大学, 2024.

⁷ Yunpeng Ma et al. “Improve thermal efficiency and reduce NOx emission of circulating fluidized bed boiler based on multi-objective optimization framework”. In: *Thermal Science and Engineering Progress* (2025), p. 103753. ISSN: 2451-9049. DOI: <https://doi.org/10.1016/j.tsep.2025.103753>. URL: <https://www.sciencedirect.com/science/article/pii/S2451904925005438>.

- 分子设计
 - 设计具有特定属性的分子结构⁹

⁹Mario Wieser. “Learning Invariant Representations for Deep Latent Variable Models”. PhD thesis. University of Basel, 2020, Mario Wieser et al. “Inverse Learning of Symmetries”. In: *Advances in Neural Information Processing Systems*. Vol. 33, 2020, pp. 18004–18015.

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环境学

- 大气污染传播路径分析
 - 兰州市大气污染传播路径预测¹³
- 火电厂排放污染物管控
 - 火电厂氮氧化物排放浓度预测¹⁴
- 氨气污染
 - 长三角地区氨气浓度长期变化及其驱动因素¹⁵
- SO_2 排放浓度预测
 - 提出了一种 SO_2 提前预测的软测量技术¹⁶

¹³ 吴京鹏. “基于图嵌入表示的节点无特征网络链路预测研究”. 硕士学位论文. 西北师范大学, 2022.

¹⁴ 金秀章, 乔鹏, and 史德金. “基于 VMD-Bayes-Lasso 算法带误差补偿的火电厂 NO_x 浓度软测量”. In: 华北电力大学学报 (自然科学版) 52 235 (2023), pp. 117–124+142.

¹⁵ Jingkai Xue et al. "Long-term spatiotemporal variations of ammonia in the Yangtze River Delta region of China and its driving factors". In: *Journal of Environmental Sciences* 150 (2025), pp. 202–217. ISSN: 1001-0742. DOI: <https://doi.org/10.1016/j.jes.2024.02.021>. URL: <https://www.sciencedirect.com/science/article/pii/S1001074224001153>.

¹⁶ 乔鹏. “基于 CatBoost-Bayes 带误差补偿的脱硫出口 SO_2 浓度软测量研究”. 硕士学位论文. 华北电力大学, 2024.

- 土地干旱度分类
 - 考虑蒸散过程的土地干旱度分类¹⁸
- 北欧山地陆气耦合分析
 - 利用 CE 分析了芬瑟两个观测站的气象观测变量之间 CE 和解耦度量之间的关系¹⁹

¹⁹Laura Mack, Marvin Kähnert, and Norbert Pirk. “Probabilistic modelling of atmosphere-surface coupling with a copula Bayesian network”. In: *Environmental Modelling & Software* (2026). arXiv: 2509.11975 [physics.ao-ph]. URL: <https://arxiv.org/abs/2509.11975>.

CASE 4: Land-atmosphere coupling by University of Oslo

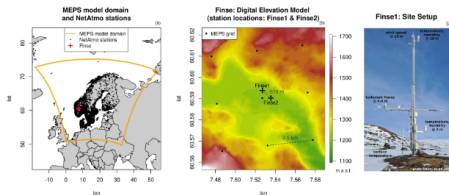


Figure 1: Used data sources: (a) Model domain of the NWP model MEPS and locations of the NetAtmo stations. (b) Surrounding of Finse and station locations. (c) Setup of the main site Finse1.

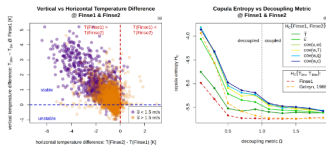


Figure 2: Relation of vertical and horizontal coupling: (a) Horizontal temperature difference between Finse1 and Finse2 versus vertical temperature difference at Finse1 colored based on a wind speed threshold (30 minutes temporal average), (b) decoupling metric at Finse1 versus copula entropy between Finse1 and Finse2 of temperature, wind speed, and the covariances corresponding to the fluxes of momentum $\text{cov}(u,w)$, sensible heat $\text{cov}(w,T)$, latent heat $\text{cov}(q,w)$ and CO_2 $\text{cov}(c,w)$ (30 minutes temporal average). Entropy calculation is based on time series of the same length to avoid a sampling size bias.

水文学 1

- 洪水预报
 - 金沙江流域洪水预报²⁰
 - 天山北麓奎屯河洪水分析²¹
- 河流相关性
 - 长江上游河段（金沙江、岷江、沱江、嘉陵江）相关性²²
 - 长江上游河段复合洪水事件分析²³
- 水沙关系分析
 - 黄河西柳沟河流域径流量和输沙量数据分析²⁴

²⁰ Lu Chen, Vijay P. Singh, and Shenglian Guo. "Measure of Correlation between River Flows Using the Copula-Entropy Method". In: *Journal of Hydrologic Engineering* 18.12 (2013), pp. 1591-1606, Lu Chen et al. "Copula entropy coupled with artificial neural network for rainfall-runoff simulation". In: *Stochastic Environmental Research and Risk Assessment* 28.7 (2014), pp. 1755-1767, 陈佳雷 et al. "一种基于多要素注意力时空图卷积网络的径流预报方法". Pat. CN111715128A, CN111715128A, 2023.

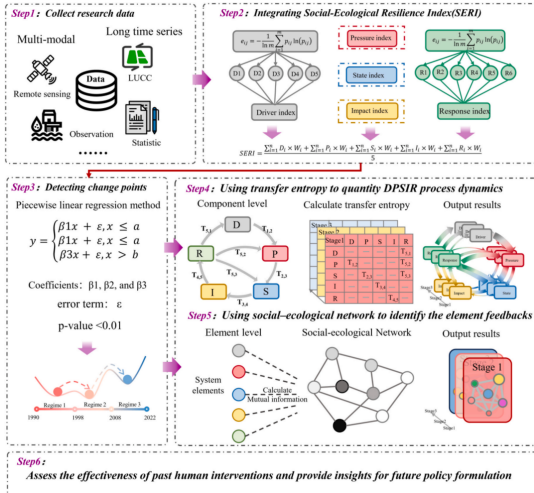
²¹Yingying Han et al. "Non-Stationary Flood Characteristics and Joint Risk Analysis in Inland China with Uncertainty Considerations". In: *Atmosphere* 17.3 (2026). p. 281.

²²Lu Chen and Shenglian Guo. *Copulas and its Application in Hydrology and Water Resources*. Springer Singapore, 2018.

²³Xu Wang and Yong-Ming Shen. "A Framework of Dependence Modeling and Evaluation System for Compound Flood Events". In: *Water Resources Research* 59.8 (2023). e2023WR034718. DOI: <https://doi.org/10.1029/2023WR034718>.

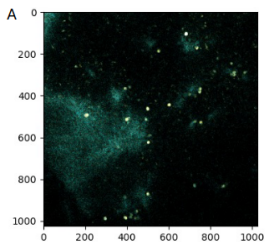
²⁴Longxia Qian et al. "A New Estimation Method for Copula Parameters for Multivariate Hydrological Frequency Analysis With Small Sample Sizes". In: *Water Resources Management* 36.4 (2022), pp. 1141–1157. ISSN: 1573-1650. URL: <https://doi.org/10.1007/s11269-021-03016-w>.

CASE 7: Social-ecological resilience in Karst region by CAS

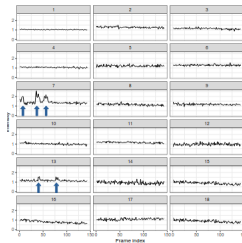


Research methodology with copula entropy based transfer entropy

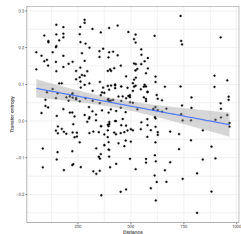
CASE 8: Intracellular calcium signals by Science Tokyo



Intravital Ca^{2+} imaging of lymphocytes



Time series of intracellular Ca^{2+} signals



Scattering plot of estimated transfer entropy

神经科学 1

● 神经病学

- 注意缺陷多动障碍⁶⁰
- 癫痫症⁶¹
- 脑卒中⁶²

⁶⁰ Paolo Victor Redondo, Raphaël Huser, and Hernando Ombao. "Measuring information transfer between nodes in a brain network through spectral transfer entropy". In: *The Annals of Applied Statistics* 19.3 (2025), pp. 2386–2411. DOI: [10.1214/25-AOAS2050](https://doi.org/10.1214/25-AOAS2050). URL: <https://doi.org/10.1214/25-AOAS2050>, 汪方毅 et al. "基于静息态 fMRI 区分健康老年人认知水平的 MVPA 方法研究". In: *磁共振成像* 14.6 (2023), pp. 18–25.

⁶¹ Zhaohui Li et al. "Estimating global phase synchronization by quantifying multivariate mutual information and detecting network structure". In: *Neural Networks* 183 (2025), p. 106984. ISSN: 0893-6080. DOI: <https://doi.org/10.1016/j.neunet.2024.106984>. URL: <https://www.sciencedirect.com/science/article/pii/S0893608024009134>.

⁶² Wojciech Ciezobka et al. "End-to-End Stroke Imaging Analysis Using Effective Connectivity and Interpretable Artificial Intelligence". In: *IEEE Access* 13 (2025), pp. 10227–10239. DOI: [10.1109/ACCESS.2025.3529179](https://doi.org/10.1109/ACCESS.2025.3529179).

神经科学 2

● 认知神经学

- 分析大脑认知活动的多模态数据⁶³
- 语音信息的编码和解析⁶⁴
- 神经元特异化识别⁶⁵

⁶³Stephanie J. Kayser et al. "Irregular Speech Rate Dissociates Auditory Cortical Entrainment, Evoked Responses, and Frontal Alpha". In: *The Journal of Neuroscience* 35.44 (2015), pp. 14691–14701, Robin A. A. Ince et al. "The Deceptively Simple N170 Reflects Network Information Processing Mechanisms Involving Visual Feature Coding and Transfer Across Hemispheres". In: *Cerebral Cortex* 26.11 (2016), pp. 4123–4135, Robin A.A. Ince et al. "A statistical framework for neuroimaging data analysis based on mutual information estimated via a gaussian copula". In: *Human Brain Mapping* 38.3 (2017), pp. 1541–1573, Etienne Combrisson et al. "Group-level inference of information-based measures for the analyses of cognitive brain networks from neurophysiological data". In: *NeuroImage* 258 (2022), p. 119347. ISSN: 1053-8119. DOI: <https://doi.org/10.1016/j.neuroimage.2022.119347>. URL: <https://www.sciencedirect.com/science/article/pii/S1053811922004669>.

⁶⁴Pieter De Clercq et al. "Beyond linear neural envelope tracking: a mutual information approach". In: *Journal of Neural Engineering* 20.2 (2023), p. 026007. DOI: 10.1088/1741-2552/acbe1d. URL: <https://dx.doi.org/10.1088/1741-2552/acbe1d>.

⁶⁵NA Pospelov et al. "Searching for cognitive specializations of neurons using mutual information framework". In: *Genes & Cells* 18.4 (2023), pp. 878–881.

神经科学 3

- 运动神经学
 - 分析运动的肌肉组合协同策略⁶⁶
- 计算神经学
 - 神经元可塑性建模⁶⁷
 - 神经网络信息传输关系分析⁶⁸

⁶⁶ 吴亚婷 et al. “多尺度肌间耦合网络分析”. In: 生物医学工程学杂志 38.4 (2021), pp. 742–752, Yating Wu et al. “R-Vine Copula Mutual Information for Intermuscular Coupling Analysis”. In: *Proceedings of the 11th International Conference on Computer Engineering and Networks*. 2022, pp. 526–534, David O’ Reilly and Ioannis Delis. “A network information theoretic framework to characterise muscle synergies in space and time”. In: *Journal of Neural Engineering* 19.1 (2022), p. 016031. DOI: 10.1088/1741-2552/ac5150. URL: <https://doi.org/10.1088/1741-2552/ac5150>, Shaojun Zhu et al. “Intermuscular coupling network analysis of upper limbs based on R-vine copula transfer entropy”. In: *Mathematical Biosciences and Engineering* 19.9 (2022), pp. 9437–9456, 金国美 et al. “基于小波包-Copula 互信息的肌间耦合特性”. In: 传感技术学报 35.10 (2022), pp. 1348–1353.

⁶⁷ Johannes Leugering and Gordon Pipa. “A Unifying Framework of Synaptic and Intrinsic Plasticity in Neural Populations.”. In: *Neural Computation* 30.4 (2018), pp. 945–986.

⁶⁸ Ari Pakman et al. “Estimating the Unique Information of Continuous Variables in Recurrent Networks”. In: *Advances in Neural Information Processing Systems* 34 (2021), pp. 20295–20307.

医学 2

● 临床医学 2

- 白内障术后角膜水肿风险预测⁷⁸
- 主动脉瓣置换手术射血分数分析⁷⁹
- 脑肿瘤医学影像组学诊断模型构建⁸⁰
- 基于脉搏波的高血压和糖尿病健康状态分类⁸¹

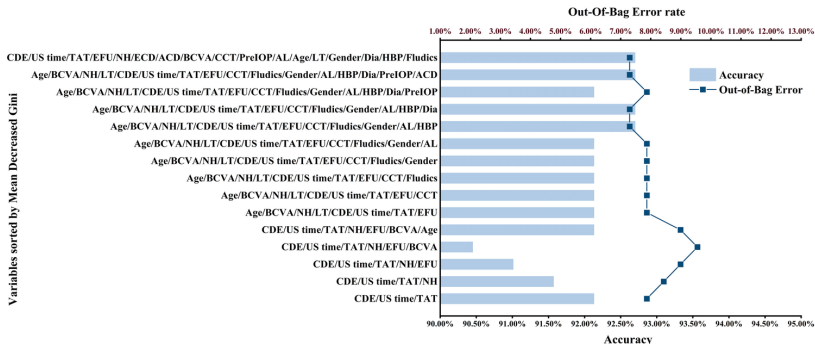
⁷⁸Yu Luo et al. "Research on Establishing Corneal Edema after Phacoemulsification Prediction Model Based on Variable Selection with Copula Entropy". In: *Journal of Clinical Medicine* 12.4 (2023), p. 1290. ISSN: 2077-0383. DOI: 10.3390/jcm12041290. URL: <https://www.mdpi.com/2077-0383/12/4/1290>.

⁷⁹S.M. Sunoj and N. Unnikrishnan Nair. "Survival Copula Entropy and Dependence in Bivariate Distributions". In: *REVSTAT-Statistical Journal* 23.1 (2025), pp. 101–115. URL: <https://revstat.ine.pt/index.php/REVSTAT/article/view/560>.

⁸⁰潘红宇. "基于影像组学与深度学习的脑肿瘤图像分类研究". 硕士学位论文. 西南大学, 2023.

⁸¹汤宇飞. "基于脉搏波的糖尿病和高血压诊断算法研究". 硕士学位论文. 中国矿业大学, 2023.

CASE 10: Corneal edema prediction by PLA 301 Hospital



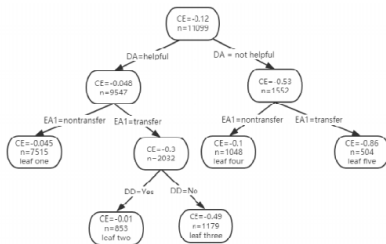
The predictive accuracy and out-of-bag error rate of each random forest model generated after selecting variables on the basis of copula entropy

药学

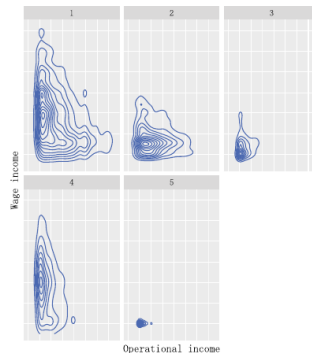
- 药物-靶点相互作用预测
 - 提出了一种交互式多特征融合的 DTIs 预测算法⁸⁷

⁸⁷ 高浩田 et al. “基于交互式多特征融合算法的药物靶标预测”. In: 太原理工大学学报 55.4 (2024), pp. 751–758.

CASE 11: Poverty policy evaluation by JXUFE



Binary decision tree by copula entropy



Joint density of two incomes of the families in each leaf

CASE 12: Commodity market analysis by UNESP



Figura 5.0.5: Transfer Entropy entre as séries da volatilidade de preços do açúcar e etanol.



Figura 5.0.6: Transfer Entropy entre as séries da volatilidade de preços do açúcar e gasolina.



Figura 5.0.7: Transfer Entropy entre as séries da volatilidade de preços do etanol e gasolina.

CASE 13: LLMs evaluation by University of Cambridge

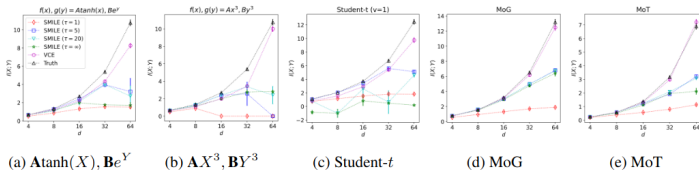


Figure 8: Comparison with the SMILE estimator under different clipping values τ .

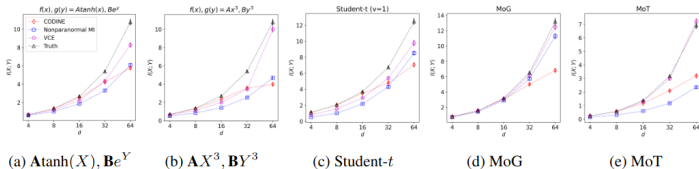


Figure 9: Comparing VCE with classic copula-based estimators e.g. nonparanormal MI (which uses a Gaussian copula to estimate MI) and CODINE (equivalent to classic copula transformation + MINE).

社会科学 4

- 法学
 - 社区属性与社区犯罪关系分析¹⁰⁵
- 政治学
 - 分析政权领导力因素和政权危机之间关系¹⁰⁶
- 军事学
 - 目标意图识别¹⁰⁷
- 情报学
 - 颠覆性技术科学-技术-产业互动模式分析¹⁰⁸
 - 高价值专利评估和发现¹⁰⁹

¹⁰⁵ Mario Wieser. "Learning Invariant Representations for Deep Latent Variable Models". PhD thesis. University of Basel, 2020.

¹⁰⁶ Stuart William Card. "Towards an Information Theoretic Framework for Evolutionary Learning". PhD thesis. Syracuse University, 2011.

¹⁰⁷ 张可 et al. "一种基于动态贝叶斯网络的目标意图识别方法". Pat. CN114997306A. CN114997306A. 2022.

¹⁰⁸ 许海云 et al. "颠覆性技术的科学-技术-产业互动模式识别与分析". In: 情报学报 42.7 (2023), pp. 816–831.

¹⁰⁹ 徐蕊婷, 陈哲宇, and 陈加鹏. "基于两阶段混合式特征选择的潜在高价值专利多情景判别方法研究". In: 数据分析与知识发现 (2026).

工学 1

● 能源工程 1

- 能源网络管理，研究天气因素与能源网络的耦合¹¹⁰
- 光伏发电功率预测¹¹¹
- 风电机组工况划分¹¹²
- 电力负荷预测¹¹³
- 风光储协同规划¹¹⁴

¹¹⁰ Xueqian Fu et al. "Uncertainty analysis of an integrated energy system based on information theory". In: *Energy* 122.122 (2017), pp. 649–662.

¹¹¹ 朱正林 和 张冕. "基于 AO 优化 VMD-CE-BiGRU 的光伏发电功率预测". In: *国外电子测量技术* 41.10 (2022), pp. 56–61.

¹¹² 崔双双 和 孙单勋. "分工况下风电机组各变量相关性研究". In: *综合智慧能源* 44.12 (2022), pp. 49–55.

¹¹³ Jian Ma. "Identifying Time Lag in Dynamical Systems with Copula Entropy based Transfer Entropy". In: *arXiv preprint arXiv:2301.06037*, arXiv:2301.06037 (2023). arXiv: 2301.06037 [cs.LG], Qin Yan et al. "Short-term prediction of integrated energy load aggregation using a bi-directional simple recurrent unit network with feature-temporal attention mechanism ensemble learning model". In: *Applied Energy* 355 (2024), p. 122159. ISSN: 0306-2619. DOI: <https://doi.org/10.1016/j.apenergy.2023.122159>. URL: <https://www.sciencedirect.com/science/article/pii/S0306261923015234>.

¹¹⁴ 董海艳 et al. "一种含源荷时序相似度约束的源储协同规划配置方法". Pat. CN114421538A. CN110766314A. 2022.

工学 2

● 能源工程 2

- 电网频率稳定性预测¹¹⁵
- 用户线损贡献分析¹¹⁶
- 配电网拓扑辨识¹¹⁷
- 电价预测¹¹⁸
- 电力系统宽频振荡影响因素和传播路径分析¹¹⁹
- 电力系统信息物理融合¹²⁰

¹¹⁵ Peili Liu et al. "Frequency Stability Prediction of Power Systems Using Vision Transformer and Copula Entropy". In: *Entropy* 24.8 (2022), p. 1165. ISSN: 1099-4300. DOI: 10.3390/e24081165. URL: <https://www.mdpi.com/1099-4300/24/8/1165>.

¹¹⁶ Wei Hu et al. "Research on User Loss Contribution Calculation of High-Loss Distribution Area Based on Transfer Entropy". In: *2022 China International Conference on Electricity Distribution (CICED)*. 2022, pp. 499–502. DOI: 10.1109/CICED56215.2022.9929052.

¹¹⁷ 秦超 and 潘毓笙. "一种基于时空特征的配电网拓扑辨识方法". Pat. CN117154679A. CN117154679A. 2023.

¹¹⁸ Xiaoping Xiong and Guohua Qing. "A hybrid day-ahead electricity price forecasting framework based on time series". In: *Energy* 264 (2022), p. 126099. ISSN: 0360-5442. DOI: <https://doi.org/10.1016/j.energy.2022.126099>. URL: <https://www.sciencedirect.com/science/article/pii/S0360544222029851>.

¹¹⁹ 冯双 et al. "一种电力系统宽频振荡影响因素和传播路径分析方法". Pat. CN114977222A. CN114977222A. 2022, 冯双 et al. "基于 Copula 传递熵的设备级和网络级宽频振荡传播路径分析及振荡源定位方法". In: *电工技术学报* 39.16 (2023), pp. 4996–5010. DOI: 10.19595/j.cnki.1000-6753.tces.230873.

¹²⁰ Fengtong Duan and Wanxing Sheng. "Conjugate Information-Energy Entropy Modeling and Entropy Gradient Control Analysis Based on Cyber-Physical Integration". In: *IEEE Transactions on Smart Grid* (2026), pp. 1–1.

CASE 14: Cyber-physical integration by State Grid

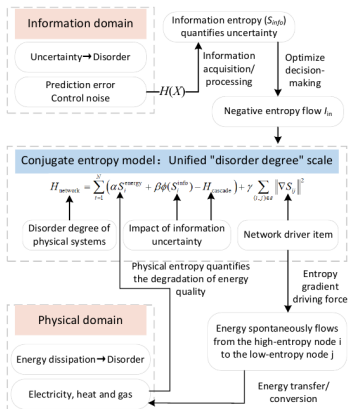


Fig. 1. The core concept diagram of the conjugate entropy theory.

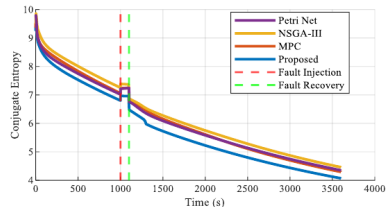


Fig. 7. Comparison of conjugate entropy.

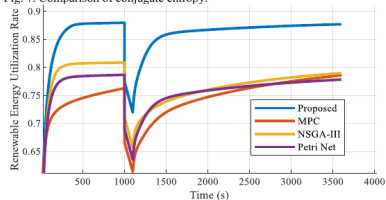


Fig. 8. Comparison of renewable energy utilization rate.

工学 4

- 纺织工程
 - 纱布生产过程质量控制¹²³
 - 可穿戴织物传感器方案设计¹²⁴
- 食品工程
 - 葡萄酒质量与理化成分关系分析¹²⁵

¹²³Seyedeh Azadeh Fallah Mortezaejad et al. "Non-Parametric Multivariate Control Chart Using Copula Entropy". In: *Sankhya B* 87 (2025), pp. 468–518. URL: <https://doi.org/10.1007/s13571-025-00374-y>.

¹²⁴Jiaqi Mo et al. "An Evaluation Framework for Textile Stretch Sensor Layouts Based on Mutual Information". In: *Companion of the 2025 ACM International Joint Conference on Pervasive and Ubiquitous Computing*. UbiComp Companion '25. Finland: Association for Computing Machinery, 2026, pp. 35–39. ISBN: 9798400714771. DOI: 10.1145/3714394.3754434. URL: <https://doi.org/10.1145/3714394.3754434>.

¹²⁵Marvin Lasserre, Régis Lebrun, and Pierre-Henri Willemin. "Learning Continuous High-Dimensional Models using Mutual Information and Copula Bayesian Networks". In: *Thirty-Fifth AAAI Conference on Artificial Intelligence, AAAI 2021*. AAAI Press, 2021, pp. 12139–12146. URL: <https://ojs.aaai.org/index.php/AAAI/article/view/17441>, Marvin Lasserre. "Apprentissages dans les réseaux bayésiens à base de copules non-paramétriques". PhD thesis. Sorbonne Université, 2022. URL: <https://tel.archives-ouvertes.fr/tel-03647090>.

CASE 15: Wine quality by Sorbonne Université

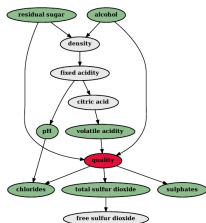


FIGURE 8.10 – Structure reconstruite avec l'algorithme B-CMIIC à partir de l'échantillon des vins rouges. Le paramètre α a été fixé à 0.04 en utilisant la méthode de *cross-validation*. La variable colorée en rouge est la variable cible tandis que l'ensemble des variables colorées en vert correspond à sa couverture de Markov.



FIGURE 8.11 – Structure reconstruite avec l'algorithme B-CMIIC pour l'échantillon des vins blancs. Le paramètre α est fixé à 0.02. La variable colorée en rouge est la variable cible tandis que l'ensemble des variables colorées en vert correspond à sa couverture de Markov.



FIGURE 8.12 – Structure reconstruite avec l'algorithme B-CMIIC pour l'échantillon des vins blancs. Le paramètre α est fixé à 0.06. La variable colorée en rouge est la variable cible tandis que l'ensemble des variables colorées en vert correspond à sa couverture de Markov.

工学 7

● 制造工程

- 制造质量管理，研究优化制造过程参数，预测产品质量¹³⁵
- 装配质量控制¹³⁶
- 工业过程故障监测¹³⁷
- 钢铁工艺过程碳排放预测¹³⁸
- 液晶显示器质量预测¹³⁹
- 航空发动机涡轮盘制造工艺优化¹⁴⁰

¹³⁵ Yan-Ning Sun et al. "Modelling and Prediction of Injection Molding Process Using Copula Entropy and Multi-Output SVR". In: *IEEE 17th International Conference on Automation Science and Engineering*. 2021, pp. 1677–1682.

¹³⁶ 王小巧. “复杂机械产品装配过程质量自适应控制方法及支持系统研究”. 博士学位论文. 合肥工业大学, 2015.

¹³⁷ Jie Dong, Keren Cao, and Kaixiang Peng. "Hierarchical Causal Graph-Based Fault Root Cause Diagnosis and Propagation Path Identification for Complex Industrial Process Monitoring". In: *IEEE Transactions on Instrumentation and Measurement* 72 (2023), pp. 1–11. DOI: 10.1109/TIM.2023.3268464, 刘鹏阳. “数据驱动的全流程分布式过程监控”. 硕士学位论文. 北方工业大学, 2023.

¹³⁸ Jie Hu et al. "Dynamic Modeling Framework Based on Automatic Identification of Operating Conditions for Sintering Carbon Consumption Prediction". In: *IEEE Transactions on Industrial Electronics* 71.3 (2023), pp. 3133–3141. DOI: 10.1109/TIE.2023.3270514.

¹³⁹ Hongxia Cai and Zhiqiang Rong. "Key Quality Feature Identification and Quality Prediction in Complex Manufacturing Processes". In: *2023 15th International Conference on Intelligent Human-Machine Systems and Cybernetics (IHMSC)*. 2023, pp. 229–232. DOI: 10.1109/IHMSC58761.2023.00060.

¹⁴⁰ 李家宝. “GH4169 合金涡轮盘产线的质量性能预测与工艺调控”. 硕士学位论文. 重庆大学, 2023.

工学 10

- 医学工程
 - 脑机接口¹⁴⁹
- 冶金工程
 - 真空蒸馏法制备高纯金属工艺参数优化¹⁵⁰

¹⁴⁹Chao Tang, Dongyao Jiang, and Badong Chen. "MEG Channel Selection Using Copula Entropy-Based Transfer Entropy for Motor Imagery BCI". In: *2024 46th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*. 2024, pp. 1–4. DOI: 10.1109/EMBC53108.2024.10782066, Chao Tang et al. "Copula Transfer Entropy-Based Channel Selection for MEG Motor Imagery Brain Computer Interfaces". In: *Tsinghua Science and Technology* 31.3 (2026), pp. 1474–1486. DOI: 10.26599/TST.2025.9010006. URL: <https://www.sciopen.com/article/10.26599/TST.2025.9010006>.

¹⁵⁰田庆华 et al. "一种真空蒸馏制备高纯金属的优化方法及优化系统". Pat. CN117577229A. CN117577229A. 2024.

工学 11

● 化学工程

- 化学过程故障监测和诊断¹⁵¹
- 化工过程因果网络构建¹⁵²
- 化工过程缺失数据补全¹⁵³
- 生物发酵过程软测量¹⁵⁴

¹⁵¹Min Yin, Jince Li, and Hongguang Li. "A CNN approach based on correlation metrics to chemical process fault classifications with limited labeled data". In: *The Canadian Journal of Chemical Engineering* 101.7 (2022), pp. 3982–3997. doi: 10.1002/cjce.24749. eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1002/cjce.24749>. URL: <https://onlinelibrary.wiley.com/doi/abs/10.1002/cjce.24749>, Yingpeng Wei and Li Wang. "Copula entropy-based PCA method and application in process monitoring". In: *2022 4th International Conference on Intelligent Information Processing (IIP)*. 2022, pp. 61–64. DOI: 10.1109/IIP57348.2022.00019, Shuangshuang Pan, Li Zhu, and Xirong Xu. "Root cause and fault propagation analysis based on causal graph in chemical processes". In: *2023 CAA Symposium on Fault Detection, Supervision and Safety for Technical Processes (SAFEPROCESS)*. 2023, pp. 1–6. DOI: 10.1109/SAFEPROCESS58597.2023.10295717.

¹⁵²Xiaotian Bi et al. "Large-scale chemical process causal discovery from big data with Transformer-based deep learning". In: *Process Safety and Environmental Protection* 173 (2023), pp. 163–177.

¹⁵³武昊. "基于深度学习的化工过程软测量建模方法研究". 博士学位论文. 北京化工大学, 2023.

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- 航空航天

- 飞行器总体参数分析和优化¹⁵⁶
- 飞行器操纵品质评估¹⁵⁷
- 卫星在轨健康状态监测¹⁵⁸
- 涡扇发动机健康状态监测¹⁵⁹
- 机场间航班延误因果关系分析¹⁶⁰
- 航班到达时间预测¹⁶¹

156 Baby Alpettiyil Krishnankutty, Rajesh Ganapathy, and Paduthol Godan Sankaran. "Non-parametric estimation of copula based mutual information". In: *Communications in Statistics - Theory and Methods* 49.6 (2020), pp. 1513–1527. DOI: 10.1080/03610926.2018.1563180. eprint: <https://doi.org/10.1080/03610926.2018.1563180>. URL: <https://doi.org/10.1080/03610926.2018.1563180>.

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158 Hao Liu et al. "Data-driven identification model for associated fault propagation path". In: *Measurement* 188 (2022), p. 110628. ISSN: 0263-2241. DOI: <https://doi.org/10.1016/j.measurement.2021.110628>. URL: <https://www.sciencedirect.com/science/article/pii/S02632242121014986>, Zefan Zeng et al. "Satellite Telemetry Data Anomaly Detection Using Causal Network and Feature-Attention-Based LSTM". In: *IEEE Transactions on Instrumentation and Measurement* 71 (2022), pp. 1–21. DOI: [10.1109/TIM.2022.3151930](https://doi.org/10.1109/TIM.2022.3151930).

¹⁵⁹ 贾如侠:“涡扇发动机故障预测及剩余寿命分析方法研究”,硕士学位论文,哈尔滨师范大学,2023.

¹⁶⁰ 吴格 et al. “一种因果关系分析方法及装置”. Pat. CN110766314A. CN110766314A. 2020.

¹⁶¹Xiaowei Tang et al. "Two Stages of Arrival Aircraft: Influencing Factors and Prediction of Integrated Arrival Time". In: *Aerospace* 12.3 (2025), p. 250. DOI: 10.3390/aerospace12030250.

¹⁰² 陈爱真 et al. “一种效能评估指标体系的约简方法和系统”. Pat. CN117634946A. CN117634946A. 2024.

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Instrumentation and Measurement 74 (2025), pp. 1–16. DOI: 10.1109/TIM.2025.3541779.

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¹⁷²Lin Liu, Cong Hu, and Xiao-Jun Wu. “CE²: A Copula Entropic Mutual Information Estimator for Enhancing Adversarial Robustness”. In: *Pattern Recognition and Computer Vision*. Springer Singapore, 2024. pp. 163–174.

工学 16

- 测绘遥感
 - 高光谱遥感数据分析¹⁷³
 - 大型工程变形体内部监测¹⁷⁴
- 海洋工程
 - 海洋底质信息探测¹⁷⁵

¹⁷³ Xuexing Zeng and T S Durrani. "Band selection for hyperspectral images using copulas-based mutual information". In: 2009 IEEE/SP 15th Workshop on Statistical Signal Processing. 2009, pp. 341–344. DOI: 10.1109/SSP.2009.5278570.

¹⁷⁴ 曹久慧 et al. "一种基于自注意力机制的变形监测方法". Pat. CN116378120A. CN116378120A. 2023, 张旭辉. "顾及邻近点相似性的机器学习变形分析方法研究". 硕士学位论文. 武汉大学, 2022.

¹⁷⁵ 赵廷. "多波束声呐海底底质智能分类关键技术研究". 博士学位论文. 哈尔滨工程大学, 2022.

金融工程 1

- 投资组合优化
 - 股票资产相关性网络分析¹⁷⁶
 - ST 股票分类¹⁷⁷
- 金融问题建模
 - Copula 函数模型选择¹⁷⁸
- 股票相关性建模
 - R-vine copula 结构建模¹⁷⁹

¹⁷⁶ Qitong Wang. "Social Networks, Asset Allocation and Portfolio Diversification". MA thesis. University of Waterloo, 2015, 廖轶楠. "基于 Copula 熵选股及集成神经网络预测的投资组合管理研究". 硕士学位论文. 南京信息工程大学, 2023.

¹⁷⁷ 朱仲儿. "多种机器学习方法的股票分类预测". 硕士学位论文. 上海师范大学, 2022.

¹⁷⁸ Rafael Calsaverini and Renato Vicente. "An information-theoretic approach to statistical dependence: Copula information". In: *EPL (Europhysics Letters)* 88.6 (2009), p. 68003. DOI: 10.1209/0295-5075/88/68003. URL: <https://doi.org/10.1209/0295-5075/88/68003>.

¹⁷⁹ Fadhah Amer Alanazi. "Truncating Regular Vine Copula Based on Mutual Information: An Efficient Parsimonious Model for High-Dimensional Data". In: *Mathematical Problems in Engineering* 2021 (2021), p. 4347957. URL: <https://doi.org/10.1155/2021/4347957>, 王念鸽. "基于互信息的 Vine Copula 模型的高频数据投资组合风险测度研究". 硕士学位论文. 浙江财经大学, 2023.

金融工程 2

- 量化金融工具箱 MLFinLab¹⁸⁰
- 金融系统性风险
 - 行业风险溢出效应分析¹⁸¹
 - 金融脆弱性度量¹⁸²
- 信用风险评价
 - 信用风险卡模型建立¹⁸³
 - P2P 借贷个人信用风险模型¹⁸⁴

¹⁸⁰ Hudson and Thames. *Machine Learning Financial Laboratory (MLFinLab)*. GitHub. 2021. URL: <https://github.com/hudson-and-thames/mlfinlab>.

¹⁸¹ 熊靖宇. “基于 Copula 熵的行业风险溢出效应分析”. 硕士学位论文. 东北财经大学, 2020.

¹⁸² Wanfang Chen and Marc G Genton. “Are you all normal? It depends!” In: *International Statistical Review* 91.1 (2023), pp. 114–139.

¹⁸³ 孔祥永 et al. “一种自动化特征工程信用风险评价系统及方法”. Pat. CN114049198A. CN114049198A. 2021.

¹⁸⁴ 彭翊庭. “个人信用风险评估模型比较——基于 Copula 熵的特征选择”. 硕士学位论文. 清华大学, 2022.

金融工程 3

- 金融产品价格预测
 - 基于因果关系的迁移学习价格预测模型构建¹⁸⁵
- 疫情对金融市场的影响
 - 疫情对股票指数相关性的影响¹⁸⁶
- 金融市场预测
 - 加密货币市场预测¹⁸⁷

¹⁸⁵Dabin Zhang et al. "A novel deep transfer learning framework with adversarial domain adaptation: application to financial time-series forecasting". In: *Neural Computing and Applications* 35 (2023), pp. 24037–24054. URL: <https://doi.org/10.1007/s00521-023-09047-1>.

¹⁸⁶Henryk Gurgul and Robert Syrek. "Mutual information between Polish subindexes –the use of copula entropy around the time of the COVID-19 pandemic". In: *Statistics in Transition new series* 25.1 (2024), pp. 23–41. DOI: 10.59170/stattrans-2024-002.

¹⁸⁷Amanda Mahmutovic. "Forecasting Cryptocurrency Returns In Adaptive Markets With an Extended Copula Based Feature Selection and Extended Copula Divergence Hybrid Loss Function". MA thesis. Linnaeus University, 2025.

Softwares

Third-party

- R: `cylcop`¹⁸⁹ (by University of Zürich and University of Minnesota);
- Python: `MLFinLab`¹⁹⁰ and `ArbitrageLab`¹⁹¹ (authored by Prof. Marcos López de Prado from Cornell University, Github 4500+ and 670+ stars);

¹⁸⁹Florian H. Hodel and John R. Fieberg. "cylcop: An R Package for Circular-Linear Copulae with Angular Symmetry". In: *bioRxiv* (2021), p. 2021.07.14.452253, Florian Hodel. *cylcop: Circular-Linear Copulas with Angular Symmetry for Movement Data*. CRAN. R package version 0.2.0, URL: <https://cran.r-project.org/package=cylcop>. 2022.

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Softwares

Third-party

- Python: **gcmi**¹⁹² (by Glasgow University, Radboud University Nijmegen, University of Amsterdam, Kyiv AI group, Github 100+ stars), **pytorch-mighty**¹⁹³ (by Kyiv AI group), **HOI**¹⁹⁴ and **Frites**¹⁹⁵ (by CNRS, INT, and Aix-Marseille Université), **Tensorpac**¹⁹⁶ (by Université de Montréal, INT, and Aix-Marseille Université), **driada**¹⁹⁷ (by Moscow State University);

¹⁹²Robin A.A. Ince et al. "A statistical framework for neuroimaging data analysis based on mutual information estimated via a gaussian copula". In: *Human Brain Mapping* 38.3 (2017), pp. 1541–1573, Robin A.A. Ince et al. *gcmi* : *Gaussian-Copula Mutual Information*. GitHub. URL: <https://github.com/robinince/gcmi>. 2020.

¹⁹³Danylo Ulianych. *The pytorch-mighty package in Python*. GitHub. 2023. URL: <https://github.com/dizcza/pytorch-mighty>.

¹⁹⁴BraiNets. *The HOI package in Python*. Github. 2024. URL: <https://brainets.github.io/hoi/>.

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¹⁹⁷Institute for Advanced Brain Studies. *driada: Dimensionality Reduction for Integrated Activity Data*. Github. URL: <https://github.com/iabs-neuro/driada>. 2024.

Softwares

Third-party

- Python: CopulaGP¹⁹⁸ (by University of Edinburgh), **Polars-ds**¹⁹⁹ (by New York University, 1.45+ million downloads, Github 650+ stars), **copul**²⁰⁰ (by Albert-Ludwigs-Universität Freiburg), **rcopula**²⁰¹, and **pycopcor**²⁰² (by Technische Universität Dresden);

¹⁹⁸Nina Kudryashova et al. "Parametric Copula-GP model for analyzing multidimensional neuronal and behavioral relationships". In: *PLoS computational biology* 18.1 (2022), e1009799, Nina Kudryashova. *Parametric Copula-GP framework*. Github. URL: <https://github.com/NinelK/CopulaGP>. 2022.

¹⁹⁹Tianren Qin. *Polars-ds: Polars Extension for General Data Science Use*. Github. URL: https://github.com/abstractqqq/polars_ds_extension. 2024.

²⁰⁰Marcus Rockel. *The copul package in Python*. Github. URL: <https://github.com/Corrram/copul>. 2026.

²⁰¹Christopher Bell. *The rcopula package in Python*. Github. URL: <https://github.com/Chrisebell124/python-copula>. 2026.

²⁰²Andreas Gocht-Zech. *The pycopcor package in Python*. Github. URL: <https://github.com/AndreasGocht/pycopcor>. 2023.

- Julia: GenEnt_11²⁰³ (the University of Edinburgh) ConcolityTools_11²⁰⁴

²⁰⁹The CE estimation method is decomposed into two functions for empirical copula estimation and entropy estimation.

Monograph

Jian Ma. "Copula Entropy: Theory and Applications". In: *arXiv preprint arXiv:2025.18168* (2025)

Thanks!

Q&A